23. VLANs and Trunking Concepts

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Phase: Phase 7 — L2, switching, and local nets
Estimated time: 1–2 days
Slug: `23-vlans`

Learning objectives

- Explain VLAN segmentation
- Understand access vs trunk ports
- Know 802.1Q tagging idea
- See VLAN subinterfaces on Linux

Prerequisites

Complete prior lessons. Prefer a Linux VM for `ip`/`ss`/`nft` examples; macOS works for many `curl`/`dig` drills.

```
# Linux VLAN subinterface example (lab)
sudo ip link add link eth0 name eth0.20 type vlan id 20
sudo ip addr add 10.20.0.10/24 dev eth0.20
sudo ip link set eth0.20 up
ip -d link show eth0.20
```

VLANs split broadcast domains — security/ops boundary, not magic encryption.

Common mistakes

- Using VLANs as only security control
- Native VLAN mismatches
- Overlapping IP plans across VLANs without routing

Exercises

Work in `exercises/23-vlans/`:

```
./test.sh
```

- **A:** make tests pass
- **B:** run 2–3 real commands from this lesson in your lab and note outputs
- **C:** rewrite answers from memory next day

Do not open `solutions/23-vlans/` until after an honest attempt.

Self-check

1. Access vs trunk?
2. 802.1Q does what?
3. Do VLANs encrypt?

Further reading

- https://www.rfc-editor.org/rfc/rfc3069